Computational PDEs — Day 2 classwork

Exercises (about 25–30 minutes)

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Format

Two questions. Work in pairs if you like.
Write the analytic answers on this sheet.

1. **Holomorphic or not** — Cauchy–Riemann by hand.
2. **Disk to a channel** — why $z = \dfrac{1}{\pi \mathrm{i}} \log \dfrac{1+\zeta}{1-\zeta}$ sends the unit circle to the walls of a strip.

1 Question 1 — Is f holomorphic?

Write $z = x + \mathrm{i}y$ and $f = u + \mathrm{i}v$. The Cauchy–Riemann equations are

$$u_x = v_y, \qquad u_y = -v_x. \tag{1}$$

A function that satisfies them (away from singularities) is holomorphic.

(a) Analytic check

For each function, decide whether Cauchy–Riemann holds. Real and imaginary parts:

$$\begin{aligned} z^3 + \sin z: &\quad u = x^3 - 3xy^2 + \sin x \, \cosh y, \quad v = 3x^2 y - y^3 + \cos x \, \sinh y, \\ z\,|z|^2: &\quad u = x(x^2+y^2), \quad v = y(x^2+y^2). \end{aligned}$$

1. $f(z) = z^3 + \sin z$. It is built from z only (z^3 and $\sin z$ are holomorphic, and so is their sum). State why $\overline{\partial} f = 0$.
2. $f(z, \overline{z}) = z\,|z|^2 = z^2\,\overline{z}$. From the formulae for u and v, compute $u_x - v_y$ (it is not identically zero).

Same test as on the slides

If f can be written using z alone, it is holomorphic. If $\overline{z}$ remains, it is not.

Your answers (Q1a):

$z^3 + \sin z$: CR holds? _____ why: _____

$z\,|z|^2$: CR holds? _____ $u_x - v_y =$ _____

2 Question 2 — Unit disk to a channel

The map used in class is

$$z = g(\zeta) = \frac{1}{\pi \mathrm{i}} \log\left(\frac{1+\zeta}{1-\zeta}\right), \qquad |\zeta| < 1. \tag{2}$$

The factor $1/\mathrm{i}$ rotates the strip by $-\pi/2$, so the channel is vertical: the walls are the lines $\operatorname{Re} z = \pm\frac{1}{2}$. (Without the i, the same logarithm gives the horizontal strip $\operatorname{Im} z = \pm\frac{1}{2}$.)

1. Let $|\zeta| = 1$ and $\zeta \neq 1$. Show that $(1+\zeta)/(1-\zeta)$ is **purely imaginary**. (Write $\zeta = e^{\mathrm{i}\theta}$ with $\theta \in (-\pi, \pi) \setminus \{0\}$, or use $\overline{\zeta} = 1/\zeta$.)
2. A purely imaginary number has argument $\pm\pi/2$. Conclude that, on the unit circle (except the point $\zeta = 1$),

$$\operatorname{Re} g(\zeta) = \pm\frac{1}{2}. \tag{3}$$

   The upper arc and the lower arc are the two vertical walls.
3. Check the centre: $g(0) = 0$. Optional: the inverse is $\zeta = (e^{\pi \mathrm{i} z} - 1)/(e^{\pi \mathrm{i} z} + 1)$. Verify it at $z = 0$.

Q2 — short derivation (part 1):

Which arc gives $\operatorname{Re} z = +1/2$**?** _____

Time box

About 25–30 minutes. If time is short, finish Q1(a) and Q2 part 1, then Q2 part 2.